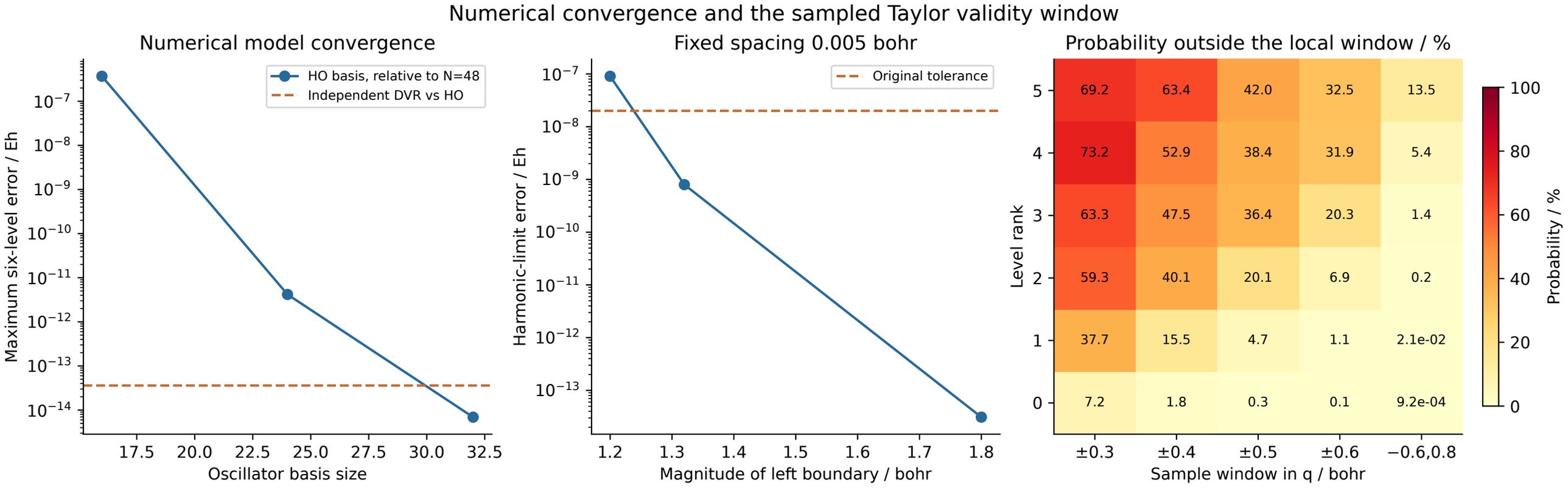
Numerical convergence and the sampled Taylor validity window
Numerical model convergence
HO basis, relative to N=48
Independent DVR vs HO
Maximum six-level error / Eh
Oscillator basis size
Fixed spacing 0.005 bohr
Original tolerance
Harmonic-limit error / Eh
Magnitude of left boundary / bohr
Probability outside the local window / %
Level rank
Sample window in q / bohr
±0.3
±0.4
±0.5
±0.6
−0.6,0.8
69.2
63.4
42.0
32.5
13.5
73.2
52.9
38.4
31.9
5.4
63.3
47.5
36.4
20.3
1.4
59.3
40.1
20.1
6.9
0.2
37.7
15.5
4.7
1.1
2.1e-02
7.2
1.8
0.3
0.1
9.2e-04
Probability / %